3-cm-BAND LOOP–GAP RESONATOR FOR SPIN-ECHO SPECTROMETER AND LASER EPR SPECTROMETER

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UDC 548.55

A 3-cm-band single-loop–single-gap reflex resonator for low-temperature studies by electron-spin-echo and magneto-optics methods is described. The resonator, which is placed in a nonresonant shield, has variable coupling with the waveguide line and provides for tuning. Tests of the resonator show that an optimum electron-spin-echo signal is achieved at a microwave power that is lower by 16 dB than that for a standard rectangular H_{102} resonator; the echo amplitude in this case is increased by 23 dB.

Loop–gap resonators, which were developed by Froncisz and Hyde [1] for 3-cm-band (≈9-GHz) EPR spectroscopy, are in many respects greatly superior to conventional resonators. The configuration of loop resonators, which are also called "split-ring" and "slot-and-hole," has essentially long been known and used in magnetrons [2]. But such resonators were first used comparatively recently in radio spectroscopy at 200 MHz to 2 GHz [3].

Features of these resonators include high density of the microwave field H_1 for a given microwave power, a high duty factor, high uniformity of H_1, relatively low Q (100-1000), and small dimensions. In addition, the electric and magnetic fields in these resonators are well-separated in space. This gave rise to the phrase "lumped-mode resonator" [1]. The listed qualities of loop resonators make possible their successful use in EPR spectrometers and electron-spin-echo (ESE) apparatuses [1, 4-6].

Various types of loop–gap resonators have been developed: from single-loop–single-gap to multiloop–multigap [7]. Advantages of the former types are simplicity of fabrication and small dimensions.

As a rule, 3-cm-band loop resonators are excited by the loop, and the microwave power is delivered through a coaxial cable. Exceptions are designs [5, 8] in which a loop resonator is placed inside a rectangular resonator. However, their effective duty factor is considerably lower than that of loop resonators in a nonresonant shield (i.e., a shield that is a supercritical waveguide for a given wavelength band).

The described resonator has two features: the presence of variable Gordon coupling between the waveguide line and the single-loop–single-gap resonator, which is placed in a nonresonant shield; the frequency is tuned by a dielectric rod. The resonator is designed for low-temperature studies on an ESE apparatus and a laser EPR spectrometer.

The resonator was designed with the aid of a semiempirical formula [9] for the resonant frequency as a function of the geometrical dimensions of a single-loop resonator (within a 10% error):

$$\nu = \frac{c}{2\pi}\left[1+\frac{r^2}{R^2-(r+w)^2}\right]^{1/2}\left[\frac{t}{\pi w \varepsilon}\right]^{1/2}\frac{1}{r}\left[\frac{1}{1+\frac{3t}{w}}\right]^{1/2}\left[1+0{,}18\,\frac{R}{z}\right]^{1/2},$$

where c is the velocity of light, R and r (in mm) are the inside radii of the shield and resonator, t (in mm) is the gap size, w and z (in mm) are the resonator thickness and length, and ε is the dielectric constant.

A diagram of the resonator is provided in Fig. 1. The microwave power is delivered through a waveguide 1 with a cross-section of 23 × 10 mm. The Gordon coupling [10] consists of a supercritical waveguide 2 with a cross-section of 12 × 10 mm with a movable Teflon wedge 3. The loop resonator 4 is located in a nonresonant cylindrical shield 5. The resonator is cemented to a support 6 of fused quartz. A plate 7 of fused quartz or leucosapphire that can be moved along the resonator gap is used for tuning. The plate 7 is attached to a movable piston 8, which is moved by a rod 9, which is soldered to a sleeve 10. Coaxial

Kazan' University. Translated from Pribory i Tekhnika Éksperimenta, No. 5, pp. 113-115, September-October, 1991. Original article submitted October 31, 1990.

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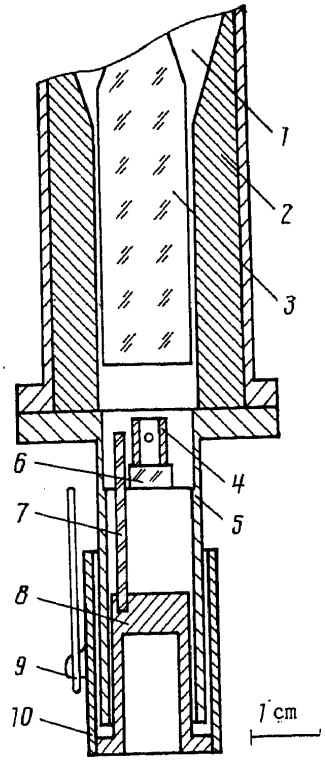


Fig. 1. Loop—gap resonator with variable coupling.

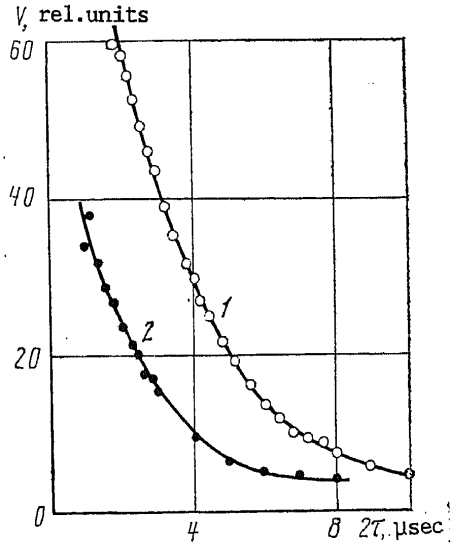


Fig. 2. Kinetics of decay of two-pulse echo of radiation centers in quartz $V = V_0 \exp(-2.8/2\tau)$: 1) loop resonator, input attenuator of receiver set to 14 dB, $V_0 = 122$; 2) rectangular resonator, input resonator completely off (0), $V_0 = 43$.

openings with diameters of 2 mm and 1 mm, respectively, are drilled in the shield and resonator perpendicular to the plane passing through the gap and axis of the resonator for optical access to the specimen. These openings increase the resonator frequency and reduce its Q slightly.

The resonator is oriented so that the gap is aimed at the narrow wall of the waveguide. The design shown in Fig. 1 was used in experiments in which it was required that the constant magnetic field H_0 and microwave field H_1 be mutually perpendicular. When H_0 must be parallel to H_1, the resonator is attached so that its axis is perpendicular to the waveguide axis.

The resonator and shield are made of brass. The inside and outside diameters of the resonator are 3 mm and 4.4 mm, the gap is 0.6 mm wide and 5 mm long, and the inside and outside diameters of the shield are 11 mm and 12 mm.

The resonator's characteristics were measured by an R2-91 panoramic VSWR meter. The loaded Q of the resonator was ≈500. The resonant frequency $\nu = 9.83$ GHz. The quartz plate ($\varepsilon \approx 3.7$) and leucosapphire plate ($\varepsilon \approx 11$) allow the resonator frequency to be lowered to 9.00 GHz and 8.80 GHz, respectively.

The loop resonator was tested in comparison with a standard rectangular H_{102} resonator on an ESE apparatus with an operating frequency of 9.4 GHz. The transmitter of the ESE spectrometer was a 1-kW magnetron with a microwave-pulse duration of 100 nsec; the receiver employed direct amplification by three traveling-wave tubes. The loaded Q of the rectangular resonator with supercritical coupling was ≈500.

The specimens were of irradiated quartz with a spin concentration of $\approx 10^{17}$ cm^{-3}, which allowed the measurements to be performed at room temperature. The decay kinetics of a two-pulse (120° + 120°) echo was studied under optimum conditions, i.e., the maximum echo signal was recorded at the lowest required microwave-pulse power.

The experiment results are presented in Fig. 2. Curve 1 corresponds to the kinetics of echo decay in the irradiated quartz using the loop resonator. In this case, the output attenuator of the magnetron transmitter was set to 23.5 dB, while the input attenuator of the receiver was set to 14 dB. Curve 2 was obtained with the rectangular resonator. The corresponding transmitter output attenuation was 7.5 dB, while the receiver input attenuator was completely off. Thus, an optimum echo signal is achieved in the loop resonator at a microwave-pulse power that is lower by 16 dB than that in the case of a rectangular H_{102} resonator. In addition, the amplitude of the echo signal in the loop resonator rose by 23 dB, owing to the higher duty factor and the better uniformity of the microwave field H_1 in the specimen.

The loop resonator was also tested on a laser EPR spectrometer [11]. The operating frequency of the spectrometer was 9.4 GHz. The specimens were phosphate glasses with impurities of rare-earth ions. The EPR signal in the loop resonator was stronger by 10 dB, as compared with the stripline resonator used in our apparatus, due to the higher uniformity and density of H_1 for the loop resonator.

The described resonator is designed for the study of disordered solids. The development of loop resonators with specimen rotation will greatly expand the range of possible specimens.

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